Physical Cume
October 19, 2007

I. Basic Thermodynamics

A **1 kg mass of ice**, originally at **-50 °C** (223 K), is heated at a constant pressure of 1 atm, until it becomes superheated steam with a temperature of **150 °C** (423 K), using an electric heater with **1 kW** power.

To simplify the calculations, we assume that

(a) The temperature is uniform through the whole mass at all times (this may be difficult to achieve for the ice but presents no problem for liquid water and steam with vigorous stirring).
(b) The specific (per kg) **heat capacity of solid ice** is **2.00 kJ K^{-1} kg^{-1}**, independent of the temperature
(c) The (specific) **heat of fusion of ice** is **333 kJ/kg**
(d) The specific **heat capacity of liquid water** is **4.18 kJ K^{-1} kg^{-1}**, independent of the temperature
(e) The (specific) **heat of vaporization of water** is **2256 kJ/kg** at 1 atm.
(f) The specific **heat capacity of steam** at 1 atm pressure is **2.08 kJ K^{-1} kg^{-1}**, independent of the temperature

1. Calculate the time (in seconds), **from the beginning of the heating** until
(A) the ice begins to melt (the first drop of water appears)
(B) the ice has melted completely (the last chunk of ice has disappeared)
(C) the water begins to boil
(D) the water has been boiled off (fully transformed to steam)
(E) the steam reaches the final temperature (150 °C)
Note: Times are cumulative, i.e. include all previous times. **(35 points)**

2. Plot the temperature of the system (in degrees Celsius) as a function of time by hand. Make sure that the graph is qualitatively correct, i.e. it shows a sharp kink if this is what happens in reality, and it is smooth if this is how the temperature behaves in reality. Only graphs that are **qualitatively correct will earn a score** in this part. **(35 points)**

3. Calculate the change of entropy from ice at -50 °C to liquid water at 0 °C, in J/K units. **(30 points)**